

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Code of Conduct

I Abinaya Sri J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

SMART HEALTHCARE APPOINTMENT AND PATIENT MANAGEMENT SYSTEM

1. Understand the Problem Statement

- **Explain the problem in your own words.**

In many hospitals and clinics, patients still face difficulties when booking appointments and managing their medical visits. They may need to call the hospital, wait in queues, or visit the hospital directly to get an appointment. During busy hours, this can cause delays, confusion, and overcrowding.

 - ✓ Patients may also find it difficult to know the available doctor, appointment time, consultation status, and previous visit details. Hospital staff have to manage many patient records and appointments manually, which can lead to errors and duplicate bookings.
 - ✓ Doctors may not get updated information about their daily appointments and patient schedules. Because of these problems, both patients and hospital staff may waste time.
 - ✓ Therefore, there is a need for a software system that can manage patient registration, doctor availability, appointment booking, and patient records in one place. The proposed system will make the appointment process easier, faster, and more organized.
- **Identify the objectives of the proposed system.**
 - ❖ To provide an easy online method for patients to register and book appointments.
 - ❖ To allow patients to search for doctors based on department and availability.
 - ❖ To show available appointment dates and time slots.
 - ❖ To reduce waiting time and overcrowding in hospitals.
 - ❖ To avoid duplicate and incorrect appointment bookings.
 - ❖ To allow patients to cancel or reschedule appointments easily.
 - ❖ To help doctors view their daily appointment schedules.
 - ❖ To maintain patient and appointment information in a centralized database.
 - ❖ To help hospital administrators manage doctors, patients, and appointments.
 - ❖ To send appointment confirmation and reminder notifications.
 - ❖ To generate useful reports for hospital management.
 - ❖ To improve the overall efficiency of hospital appointment management.

- ❖ To provide a simple and user-friendly interface for patients and staff.
- ❖ To protect patient information through proper security mechanisms.
- **State the scope of the problem.**
 - ☆ The scope of the proposed system mainly focuses on managing hospital appointments and basic patient information through a web or mobile-based application.
 - ☆ Patients can register, log in, search for doctors, check availability, and book appointments. They can also cancel or reschedule their appointments when required.
 - ☆ Doctors can view their appointment schedules and update the consultation status of patients. Hospital administrators can manage patient details, doctor information, departments, appointment slots, and system reports.
 - ☆ The system can store appointment history and basic patient information securely in a centralized database. Notifications can be provided to remind patients about upcoming appointments.
 - ☆ The system is mainly designed for hospitals, clinics, and healthcare centres that need a simple digital solution for appointment and patient management.

2. Identify Key Stakeholders and Challenges

- **Identify all the stakeholders involved in the system.**
 - ☆ **Patients:** Use the system to register, search for doctors, and book appointments.
 - ☆ **Doctors:** View appointments and manage patient consultation details.
 - ☆ **Receptionists:** Help manage appointments and patient information.
 - ☆ **Hospital Administrators:** Manage the overall system and hospital information.
 - ☆ **Hospital Management:** Use reports for monitoring and decision-making.
 - ☆ **System Developers:** Design, develop, test, and maintain the software.
 - ☆ **System Administrators:** Manage servers, databases, backups, and security.
 - ☆ **Software Testers:** Check whether the system works correctly.
 - ☆ **Notification Service Providers:** Support SMS, email, or other appointment reminders.
 - ☆ **Database Administrators:** Maintain and manage the patient and appointment database.
- **Explain the roles and responsibilities of each stakeholder.**
 - ☆ **Patients:** Provide correct personal information and use the system for booking, cancelling, or rescheduling appointments.

- ☆ **Doctors:** Check their appointment schedule, view required patient information, and update consultation status.
 - ☆ **Receptionists:** Register patients when required, manage appointments, and help patients who face difficulties using the system.
 - ☆ **Hospital Administrators:** Manage doctors, departments, appointment slots, users, and system settings.
 - ☆ **Hospital Management:** Monitor hospital activities using reports and use the information for better planning.
 - ☆ **System Developers:** Design the software, develop the modules, connect the database, and fix software problems.
 - ☆ **System Administrators:** Maintain servers, databases, backups, user access, and overall system availability.
 - ☆ **Software Testers:** Test the application, identify errors, and make sure the system functions properly before deployment.
 - ☆ **Database Administrators:** Maintain patient and appointment data and ensure that the database is secure and available.
- **Discuss the major challenges and issues faced by the stakeholders.**
- ☆ Hospitals may have a large number of patients, making appointment management difficult.
 - ☆ Incorrect patient information may lead to problems in appointment records.
 - ☆ Double booking may happen if appointment slots are not updated properly.
 - ☆ Patients may have difficulty using the system if the interface is complicated.
 - ☆ Doctors may face problems when appointment schedules change suddenly.
 - ☆ Hospital staff need proper training to use the new system effectively.
 - ☆ Patient medical information is sensitive, so data privacy and security are important.
 - ☆ Internet or server problems may temporarily stop patients from booking appointments.
 - ☆ The system must handle many users at the same time without becoming slow.
 - ☆ Integration with SMS, email, payment, or existing hospital systems may increase system complexity.
 - ☆ Regular maintenance and software updates are required to keep the system reliable.

3. Analyse the Current Approach

- **Describe the existing system or current process.**

The existing appointment process in many hospitals is based on phone calls, reception counters, or manual records. Patients may need to call the hospital or directly visit the reception desk to ask for an appointment.

Receptionists usually check doctor availability and enter appointment information into registers or computer systems. During busy hours, many patients may have to wait for a long time.

Doctors generally receive their appointment schedule through hospital staff or existing internal systems. Patient information may be stored in different files or systems, making it difficult to access information quickly.

Some hospitals already use computerized appointment systems, but these systems may provide only basic booking facilities. They may not provide complete appointment tracking, reminders, centralized records, and useful reports.

As a result, the current process can become slow and difficult to manage when the number of patients increases.

- **Identify its strengths and limitations.**

Strengths:

- ☆ Patients can directly communicate with hospital staff.
- ☆ Receptionists can help patients who are not familiar with digital systems.
- ☆ Doctors can follow their daily appointment schedules.
- ☆ Hospitals can maintain basic patient and appointment records.
- ☆ Existing manual processes do not require advanced technical knowledge.
- ☆ Emergency or special cases can be handled directly by hospital staff.

Limitations:

- ☆ Patients may have to wait in long queues.
- ☆ Appointment booking through phone calls can take more time.
- ☆ Manual records can contain errors or missing information.
- ☆ Double booking may occur due to incorrect updates.
- ☆ Patients may not know the latest doctor availability.

- ☆ There may be no automatic appointment reminders.
- ☆ Finding previous appointment information can take time.
- ☆ Hospital staff have to spend more time managing records manually.
- ☆ The system may not provide proper reports for hospital management.
- ☆ Patient information may not be centralized.

- **Analyse the problems or gaps that need to be addressed.**

Several gaps can be seen in the existing appointment management process.

- ☆ The major problem is the lack of a centralized appointment management system.
- ☆ Patients cannot always check doctor availability in real time.
- ☆ Manual booking can result in appointment conflicts and duplicate entries.
- ☆ Patients may need to spend unnecessary time waiting at the hospital.
- ☆ There may be limited facilities for cancelling or rescheduling appointments.
- ☆ Appointment reminders are often not automated.
- ☆ Patient and appointment records may be stored separately, making information retrieval difficult.
- ☆ Hospital management may not receive proper reports about appointments and patient activities.
- ☆ The current process may not handle a large number of patients efficiently.
- ☆ Better security is required to protect sensitive patient information.

4. Propose a Suitable Solution Using Software Engineering Principles

- **Design a software-based solution for the given problem**

- ☆ The proposed solution is a Smart Healthcare Appointment and Patient Management System that provides one centralized platform for patients, doctors, hospital staff, and administrators.
- ☆ Patients can create an account, log in, search for doctors, check available appointment slots, and book appointments. They can also cancel or reschedule appointments based on availability.
- ☆ Doctors can log in and view their daily appointment schedules. They can update appointment status and manage their consultation-related activities through the system.
- ☆ Hospital administrators can manage doctors, departments, patients, appointment slots, and system settings. The system will store important information in a centralized database.

- ☆ Automatic notifications can be sent to patients when an appointment is booked, cancelled, rescheduled, or approaching. Proper authentication and access control will be used to protect sensitive patient information.
- ☆ Overall, the proposed system will reduce manual work, minimize appointment errors, improve communication, and provide a more convenient healthcare appointment experience.

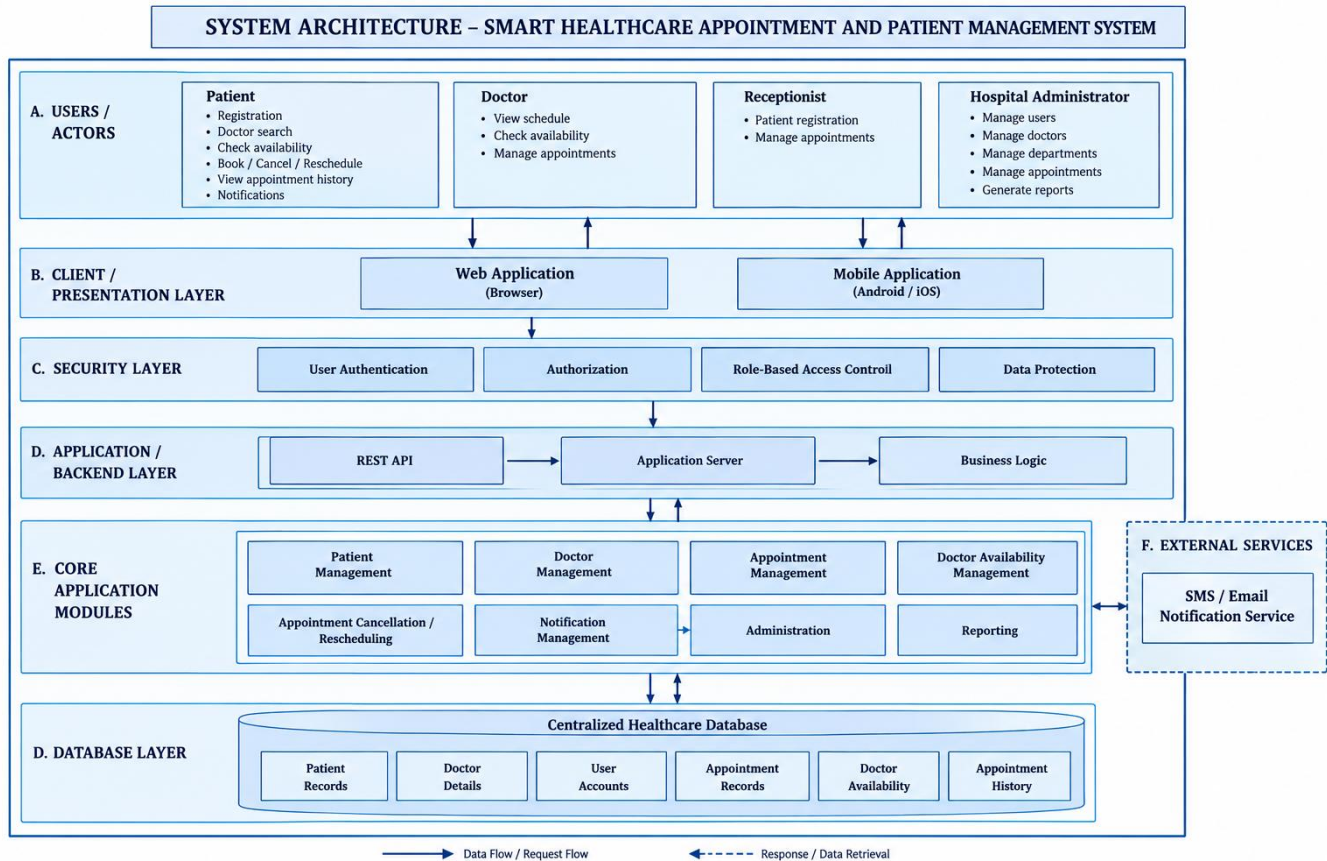


Fig 1: System Architecture of Smart Healthcare Appointment and Patient Management System

○ **Describe the major functional modules of the proposed system.**

1. Patient Management Module

This module manages patient registration, login, profile information, and appointment history. Patients can update their basic details whenever required.

2. Doctor Management Module

This module stores doctor information such as name, specialization, department, availability, and working schedule. Administrators can update doctor details when necessary.

3. Appointment Management Module

This module handles appointment booking, cancellation, and rescheduling. It checks available time slots and helps prevent duplicate bookings.

4. Consultation Management Module

Doctors can use this module to view their appointments and update the status of patient consultations.

5. Notification Module

This module sends appointment confirmation, reminder, cancellation, and rescheduling notifications through supported communication services.

6. Administration Module

Administrators can manage patients, doctors, departments, appointment slots, user accounts, and system settings.

7. Database Module

The database module securely stores patient information, doctor details, appointment records, and other system data.

8. Reporting Module

This module generates reports about appointments, patient activities, doctor schedules, cancellations, and other useful information.

9. Security and Authentication Module

This module handles login, authentication, role-based access, password protection, and secure access to sensitive information.

○ **Identify the functional and non-functional requirements.**

Functional requirements:

- ☆ The system shall allow patients to register and log in.
- ☆ Patients shall be able to update their profile information.
- ☆ The system shall allow patients to search for doctors.
- ☆ Patients shall be able to search based on department or specialization.
- ☆ The system shall display available appointment dates and time slots.
- ☆ Patients shall be able to book appointments.
- ☆ Patients shall be able to cancel or reschedule appointments.
- ☆ The system shall maintain appointment history.
- ☆ Doctors shall be able to view their appointment schedules.
- ☆ Doctors shall be able to update appointment status.

- ☆ Administrators shall be able to manage doctors and patients.
- ☆ Administrators shall be able to manage departments and appointment slots.
- ☆ The system shall send appointment confirmation notifications.
- ☆ The system shall send reminders before appointments.
- ☆ The system shall generate useful reports for hospital management.

Non-functional requirements:

- ☆ The system must provide fast response time and should remain available even when many users access it at the same time.
- ☆ The software must be scalable to support increasing numbers of patients, doctors, and appointments.
- ☆ Security mechanisms must protect patient information through secure authentication, access control, and data protection.
- ☆ The application should provide a simple and user-friendly interface so that patients and hospital staff can use it without technical knowledge.
- ☆ The system should be reliable and maintainable, with regular data backup and easy support for future updates.

- **Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.**

Modularity

- The system is divided into Patient, Doctor, Appointment, and Administration modules.
- Each module has a specific responsibility.
- Changes in one module can be made with minimum impact on other modules.

Scalability

- The system can support an increasing number of patients and doctors.
- Additional hospitals or branches can be added in future.
- Database and server resources can be expanded when required.

Maintainability

- The system is divided into independent components.
- Standard coding practices can be followed.
- Future changes can be implemented more easily.

Reliability

- Input validation reduces incorrect data.

- Database constraints help prevent duplicate appointments.
- Regular backups can reduce the risk of data loss.

Usability

- Simple navigation is provided.
- Separate dashboards can be provided for different users.
- Important appointment information is displayed clearly.

Security

- User authentication is required.
- Role-based access control restricts sensitive information.
- Passwords should be securely stored.
- Patient information should only be accessible to authorized users.

5. Justify the Chosen Methodology/Framework

- **Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).**

The Agile Software Development Life Cycle (SDLC) is selected for developing the Smart Healthcare Appointment and Patient Management System. Agile follows an iterative approach where the software is developed in small parts called sprints.

Instead of developing the complete application at one time, the development team can first create basic features such as registration and login. Later, appointment booking, doctor management, notifications, and reports can be added through separate development cycles.

Agile allows continuous communication between developers, testers, hospital staff, doctors, and other stakeholders. Their feedback can be used to improve the system after every sprint.

Healthcare requirements may change depending on hospital needs and user feedback. Agile makes it easier to handle these changes during development.

Therefore, Agile provides a flexible approach for developing, testing, improving, and maintaining the proposed healthcare system.

- **Justify why the selected methodology/framework is suitable for the proposed system.**

Agile is suitable for this project because the system contains many different modules that can be developed independently.

- ☆ Patient registration can be developed in an initial sprint.

- ☆ Doctor management can be developed in another sprint.
- ☆ Appointment booking and cancellation can be developed in the next sprint.
- ☆ Notifications and reports can be added in later sprints.
- ☆ Testing can be performed after every sprint.
- ☆ Feedback can be collected from hospital staff and users regularly.
- ☆ Changes in requirements can be added to future sprints.
- ☆ Bugs can be identified and corrected earlier.
- ☆ Developers and stakeholders can communicate throughout the development process.
- ☆ New features can be added without rebuilding the complete application.

For example, if hospital staff later request online payment or video consultation, these features can be added to the product backlog and developed in future sprints.

- **Explain how it supports successful project development and maintenance.**

- ☆ Agile divides the project into smaller tasks, making the development process easier to manage. Each sprint focuses on a particular group of features and produces a working part of the application.
- ☆ Continuous testing helps developers identify errors at an early stage. Stakeholders can review the developed features and provide feedback before the next sprint begins.
- ☆ Regular communication between developers, testers, doctors, and hospital staff helps the team understand actual user requirements. If any requirement changes, it can be included in a future sprint.
- ☆ Agile also supports maintenance after deployment. Bugs, security problems, performance issues, and new feature requirements can be handled through planned updates.
- ☆ Because the application is divided into separate modules, developers can modify one part without affecting the complete system. This makes future maintenance and improvements easier.

6. Discuss Expected Outcomes and Limitations

- **Describe the expected benefits of the proposed solution.**

The proposed Smart Healthcare Appointment and Patient Management System is expected to improve the appointment process for both patients and hospital staff.

- ☆ Patients can book appointments from home without standing in long queues.

- ☆ Doctor availability can be checked before booking.
 - ☆ Waiting time at reception counters can be reduced.
 - ☆ Duplicate appointment bookings can be minimized.
 - ☆ Patients can cancel or reschedule appointments easily.
 - ☆ Appointment reminders can reduce missed appointments.
 - ☆ Doctors can clearly view their daily schedules.
 - ☆ Receptionists can reduce repetitive manual work.
 - ☆ Patient and appointment information can be accessed more easily.
 - ☆ Hospital management can use reports for better planning.
 - ☆ Centralized data management can improve record organization.
 - ☆ Secure access can help protect patient information.
 - ☆ The system can improve communication between patients and hospital staff.
 - ☆ The overall appointment management process can become faster and more organized.
- **Explain how the solution addresses the identified challenges.**
 - ☆ The proposed system addresses the main problems of the existing appointment process by bringing different activities into one centralized platform.
 - ☆ Patients can check doctor availability and book appointments online, which reduces the need to wait at the reception counter. The appointment module can also prevent two patients from booking the same time slot.
 - ☆ Automatic notifications can remind patients about their appointments and inform them when appointments are cancelled or rescheduled. This can reduce missed appointments and communication problems.
 - ☆ Centralized patient and appointment records make it easier for authorized users to find required information. Hospital administrators can also generate reports without depending completely on manual records.
 - ☆ Security features such as authentication and role-based access can protect sensitive patient information. Regular backups can also reduce the risk of losing important data.
 - ☆ The modular design allows the system to be improved gradually as the hospital's requirements increase.
 - **Discuss potential risks, implementation challenges, and limitations.**

Although the proposed system provides many benefits, some risks and limitations may occur during implementation.

- ☆ Some patients may not be comfortable using an online system.
- ☆ Elderly patients may require assistance when booking appointments.
- ☆ Internet problems may prevent users from accessing the system.
- ☆ Server failures may temporarily stop appointment services.
- ☆ Incorrect patient information can affect the accuracy of records.
- ☆ Patient data is sensitive, so security breaches must be prevented.
- ☆ Hospital staff may require training before using the new system.
- ☆ Integrating the system with existing hospital software may be technically difficult.
- ☆ Developing and maintaining the software may require additional cost.
- ☆ The system must be properly designed to handle a large number of users.
- ☆ SMS or email services may sometimes fail due to external service problems.
- ☆ Regular maintenance and security updates are necessary.
- ☆ Data migration from old hospital records may take time.
- ☆ The system supports healthcare administration, but it cannot replace doctors or professional medical decision-making.

○ **Suggest possible future enhancements.**

The proposed system can be improved further by adding new technologies and services in future versions.

- ☆ **Mobile Application:** An Android and iOS application can be developed so that patients can manage appointments using their smartphones.
- ☆ **Online Video Consultation:** Patients can consult doctors remotely through secure video calls.
- ☆ **Online Payment:** Patients can pay consultation fees through digital payment methods.
- ☆ **Electronic Medical Records:** Medical history and previous consultation information can be integrated into the system.
- ☆ **AI-Based Doctor Recommendation:** AI can help suggest suitable doctors or departments based on the patient's selected symptoms or requirements.
- ☆ **Smart Notifications:** Personalized SMS, email, and application notifications can remind patients about appointments.
- ☆ **Multilingual Support:** Multiple languages can be added so that more patients can use the system comfortably.

- ☆ **Multi-Hospital Support:** Patients can manage appointments from different hospitals through the same platform.
- ☆ **Laboratory Integration:** Laboratory test booking and report access can be connected with the system.
- ☆ **Pharmacy Integration:** Prescription and medicine-related services can be connected in future.
- ☆ **Emergency Appointment Feature:** A separate facility can be provided for urgent appointment requests.
- ☆ **Analytics Dashboard:** Hospital management can use dashboards to understand patient visits, appointment cancellations, doctor workload, and other trends.
- ☆ **Wearable Device Integration:** Future versions may connect with wearable devices for selected health-related monitoring features.

Overall, these enhancements can help the system develop from a simple appointment application into a more complete digital healthcare management platform.

Conclusion

The Smart Healthcare Appointment and Patient Management System provides a practical software solution for improving the existing hospital appointment process. It reduces manual work, helps patients book appointments more easily, and allows doctors and administrators to manage schedules and records efficiently.

The system applies important software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security. The Agile methodology also allows the system to be developed in small stages and improved continuously through stakeholder feedback. With future enhancements such as mobile applications, online consultation, digital payments, electronic medical records, and AI-based features, the system can support more healthcare activities and provide a better experience for patients and hospital staff.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5